

Module 2

Introduction to GitHub



Agenda



GitHub Foundations Part 1

- 1 Introduction to Git
- 2 Introduction to GitHub 
- 3 Introduction to GitHub's products
- 4 Configure code scanning on GitHub
- 5 Introduction to GitHub Copilot
- 6 Code with GitHub Codespaces
- 7 Manage your work with GitHub Projects
- 8 Communicate effectively on GitHub using Markdown

GitHub Foundations Part 2

- 9 Contribute to an open-source project on GitHub
- 10 Manage an InnerSource program by using GitHub
- 11 Maintain a secure repository by using GitHub best practices
- 12 Introduction to GitHub administration
- 13 Authenticate and authorize user identities on GitHub
- 14 Manage repository changes by using pull requests on GitHub
- 15 Search and organize repository history by using GitHub
- 16 Using GitHub Copilot with Python

Introduction

In this module, you'll learn the basics of GitHub and gain a better understanding of its fundamental features with a hands-on exercise all within a GitHub repository.

By the end of this module, you'll be able to:

- Identify the fundamental features of GitHub.
- Learn about repository management.
- Gain an understanding of the GitHub flow, including branches, commits, and pull requests.
- Explore the collaborative features of GitHub by reviewing issues and discussions.
- Recognize how to manage your GitHub notifications and subscriptions.

Foundry

What is GitHub?

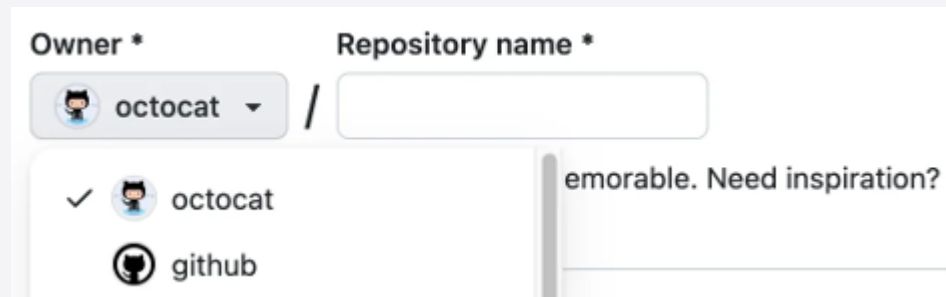
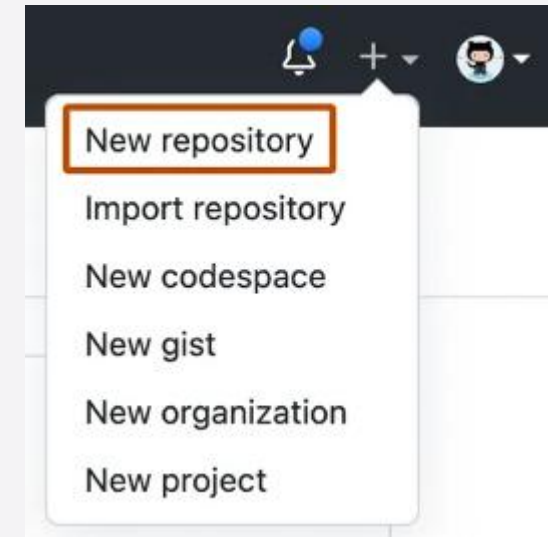
The GitHub platform simplifies the process of collaborating on projects and provides a website, command-line tools, and overall flow that allows developers and users to work together.



What is GitHub? (Introduction to repositories)

Public repositories are accessible to everyone on the internet.

Private repositories are only accessible to you, people you explicitly share access with, and, for organization repositories, certain organization members.

A screenshot of the GitHub repository creation form. It features two main sections: 'Owner *' and 'Repository name *'. The 'Owner' section has a dropdown menu with 'octocat' selected. Below it, a list of suggestions is shown, including 'octocat' (marked with a checkmark) and 'github'. The 'Repository name' section has an empty text input field. To the right of the input field, there is a link that says 'memorable. Need inspiration?'. The form is styled with a clean, modern design using GitHub's signature colors.

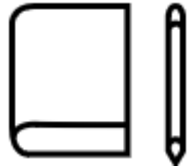
Components of the GitHub flow



Branches

*master
main*

- Make changes without affecting the entire project.
- Safe place to experiment with new features or fixes.
- Revert changes or push more changes.
- Changes won't update on the default branch until you merge your branch.



Commits

- A change to one or more files on a branch.
- Every time a commit is created, it's assigned a unique ID and tracked along with the time and contributor.
- Commits provide an audit trail for reviewing the history of a file or linked item.
- The primary states for a file in a Git repository are **Untracked** and **Tracked**.

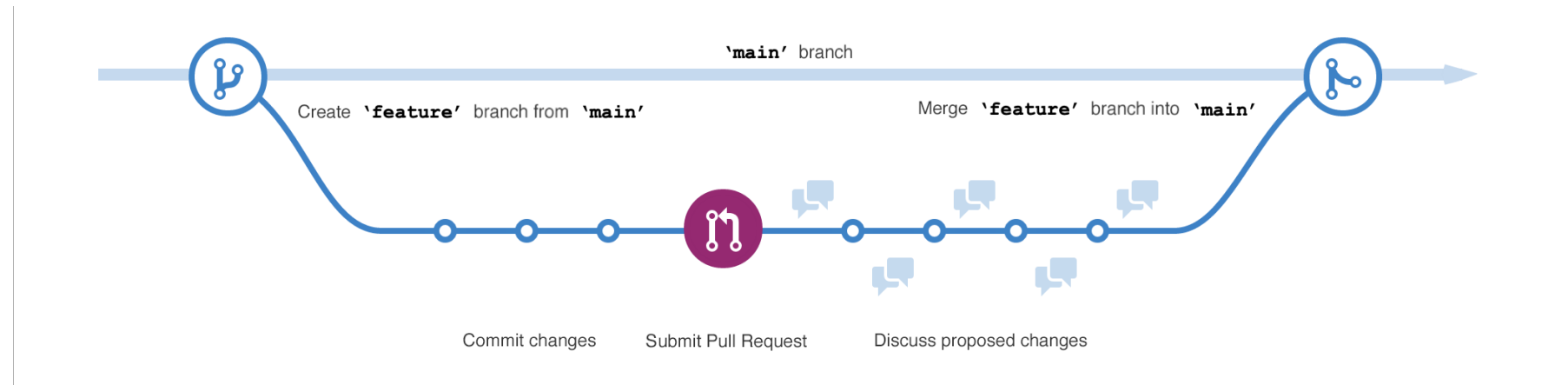


Pull requests

- Mechanism used to signal that the commits from one branch are ready to be merged into another branch.
- The team member submitting the pull request asks one or more reviewers to verify the code and approve the merge.
- Reviewers can comment on changes, add their own, or use the pull request itself for further discussion.

Components of the GitHub flow (cont.)

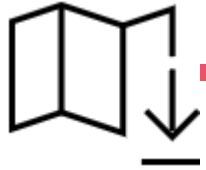
1. Create a branch so changes/features/fixes don't affect the main branch.
2. Make changes. We recommend deploying changes to your feature branch before merging into the main branch.
3. Create a pull request to ask collaborators for feedback.
4. Review and implement your feedback.
5. Get the pull request approved and merge it into the main branch.



The GitHub flow is lightweight workflow that allows for safe experimentation.

You can test new ideas and collaboration with your team by using branching, pull requests, and merging.

GitHub is a collaborative platform



Creating an Issue:

- Navigate to the repository's main page, select "Issues," then "New issue."
- Use templates or open a blank issue.
- Enter a title and description, and optionally assign, label, or add to a project.
- Submit the new issue.



Using GitHub Discussions:

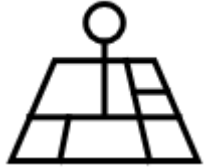
- Discussions are for open, accessible conversations not related to code.
- Enable discussions using repository settings, configure a welcome post, and start discussions.
- Discussions can be created by any user who can view the repository.



Enabling Discussions:

- Repository owners or those with Write access can enable discussions.
- Navigate to repository settings, scroll to Features, and select "Setup discussions."
- Edit the template for the welcome post and start the discussion.

GitHub platform management



Branches

- You can subscribe to notifications for specific issues, pull requests, gists, CI activity, repository events, and more.
- Automatic subscriptions occur when you open or comment on issues/pull requests, or are assigned to them.
- You can unsubscribe or customize notifications as needed.



Commits

- Notifications are updates for activities you're subscribed to.
- You can use qualifiers like mentions: to find issues mentioning a specific user.



GitHub Pages

- GitHub Pages allows you to host a static website directly from a GitHub repository.
- It supports HTML, CSS, and JavaScript, and can include a build process for publishing.

Jeckyll
Jekyll

Exercise - A guided tour of GitHub



Tour guides you through everything you need to start using GitHub in less than an hour.

This exercise checks your knowledge of GitHub key features:

- Committing a branch
- Committing a file
- Opening a pull request
- Merging a pull request



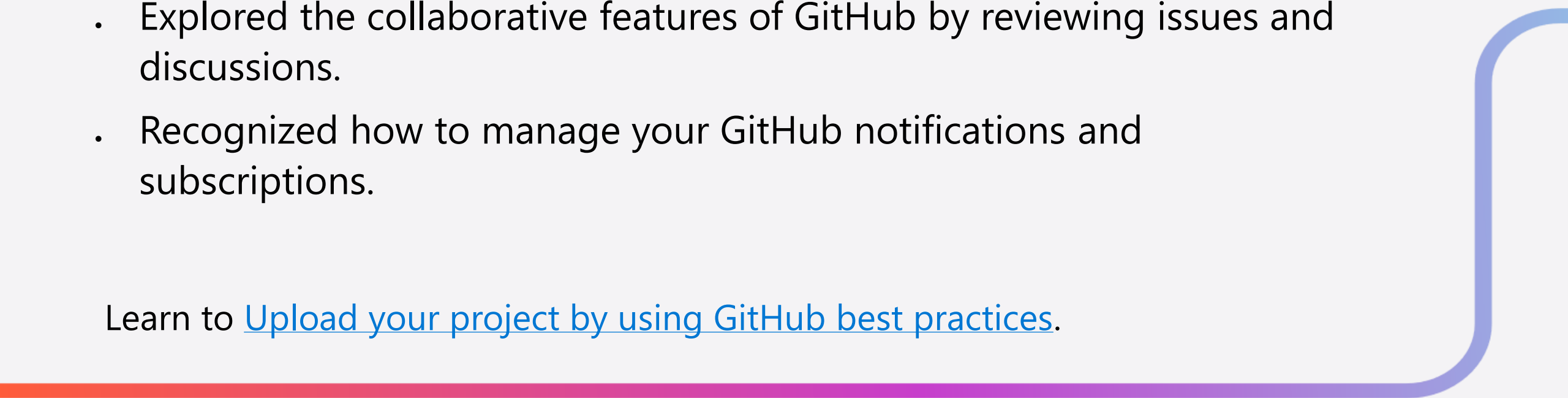
Start course

Summary

In this module, you:

- Learned about repository management.
- Gained an understanding of the GitHub flow including branches, commits, and pull requests.
- Explored the collaborative features of GitHub by reviewing issues and discussions.
- Recognized how to manage your GitHub notifications and subscriptions.

Learn to [Upload your project by using GitHub best practices](#).



End of presentation



